Name: _____

WORKSHEET

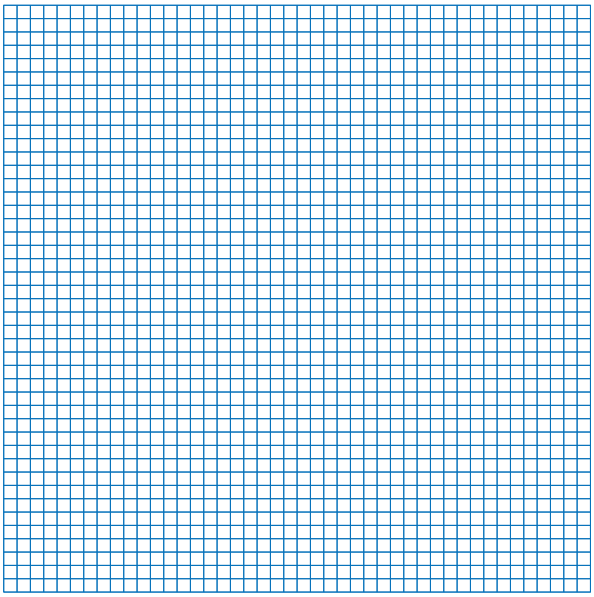
Graphs of rates of change

1 The position (in metres) of a dragster from the starting line after t seconds is given by $x = 20t^2$.

a Complete the table below.

t	0	1	2	3	4	5
x (m)						

b Graph the position of the dragster with respect to time.



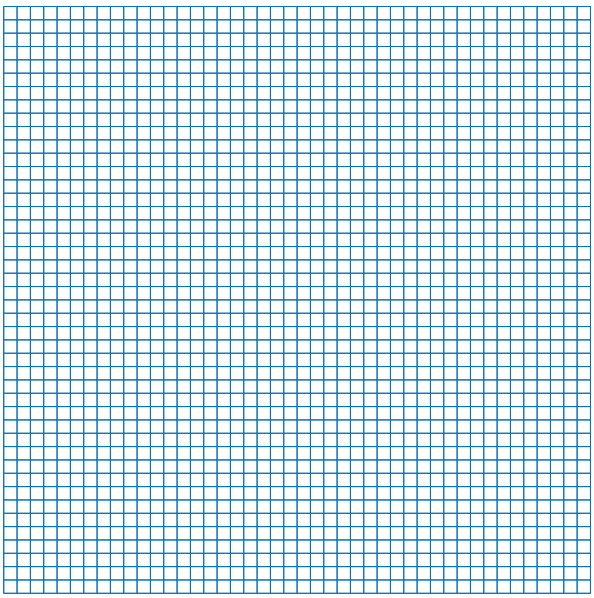
c Find the instantaneous rate of change in position (the speed v) of the dragster.

d Complete the table below.

t	0	1	2	3	4	5
v (m/s)						

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e Graph the velocity of the dragster against time.

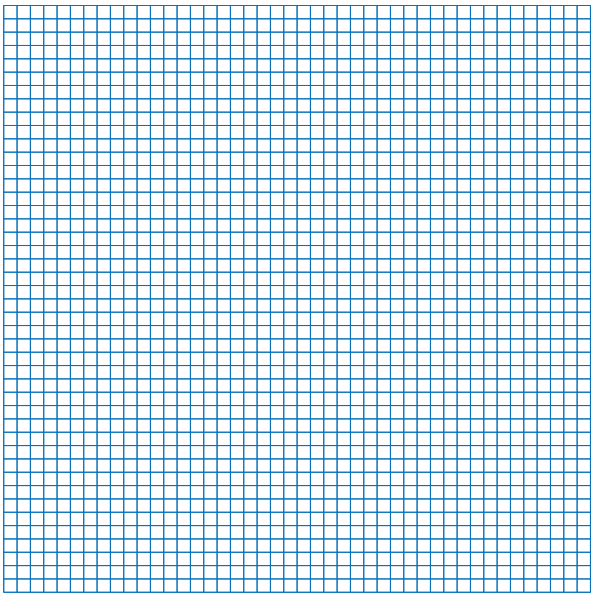


f Find the instantaneous rate of change in speed (acceleration a) of the dragster.

g Complete the table below.

t	0	1	2	3	4	5
a (m/s^2)						

h Graph the acceleration of the dragster against time.



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2 The position (in metres) of a 100 m sprinter is given by $x = 2t^2$ during the first 4 seconds of a race. After this time, the sprinter completes the race at a constant speed.

a Find an expression for the sprinter's instantaneous rate of change in position (speed) during the first 4 seconds of the race.

b What is their speed at $t = 2s$?

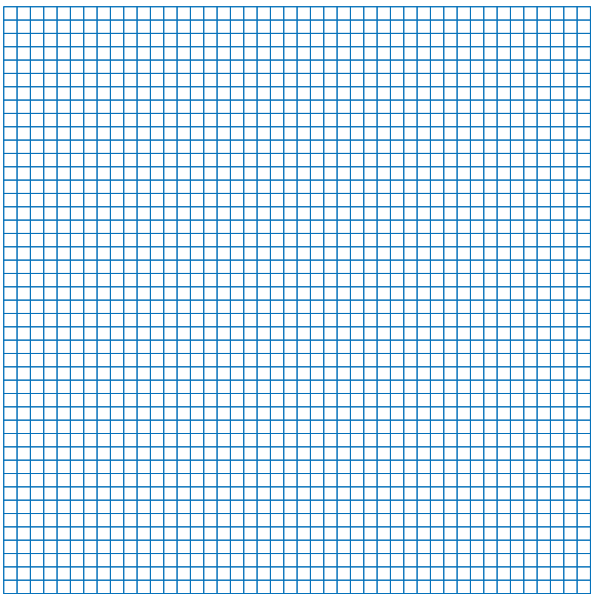
c What is their speed at $t = 4s$?

d Find an expression for the acceleration of the sprinter during the first 4 seconds.

e What is the acceleration at $t = 3$s?

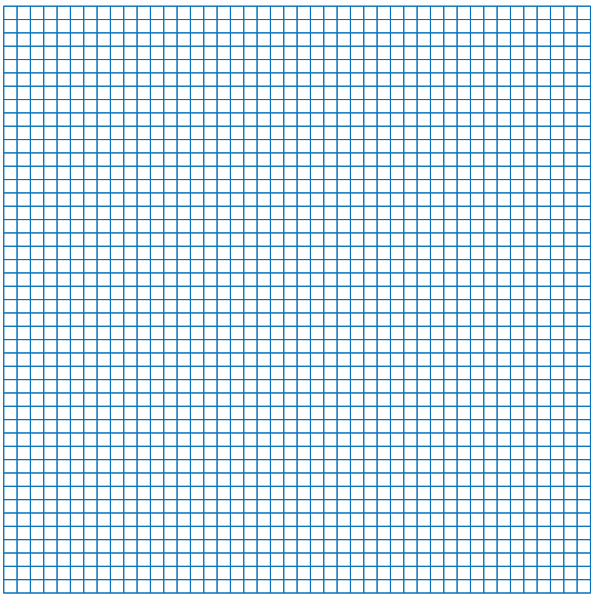
f What is the acceleration after $t = 4$s?

g Draw a distance–time graph.

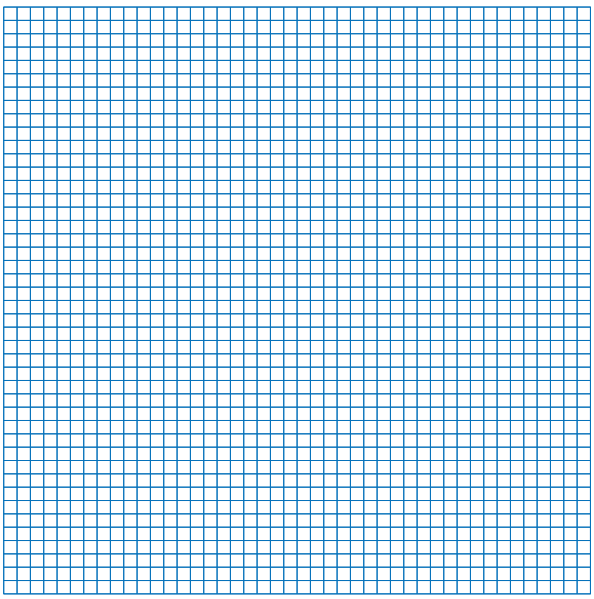


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h Draw a speed–time graph.



i Draw an acceleration–time graph



3 A stone is dropped from a 45m tower. The position (metres) relative to top of the tower after t seconds is given by $x = -5t^2$.

a Find an expression for the speed of the stone.

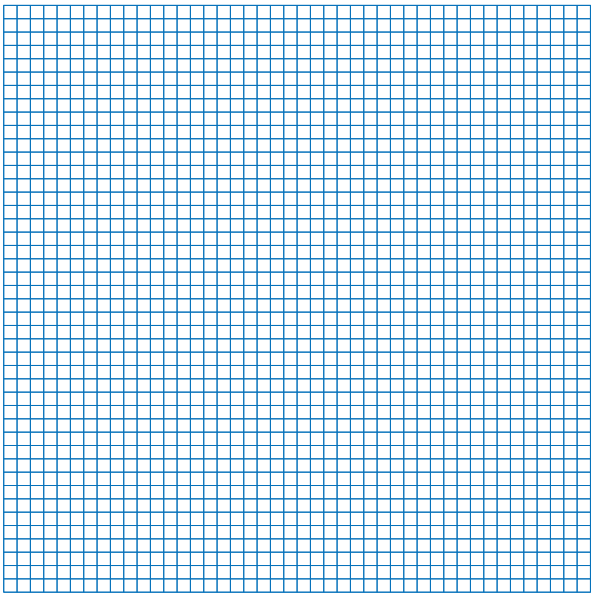
b Find an expression for the acceleration of the stone.

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c Complete the table below.

t	0	0.5	1	1.5	2	2.5	3
x (m)							
v (m/s)							
a (m/s^2)							

d Draw graphs showing position, speed and acceleration against time.



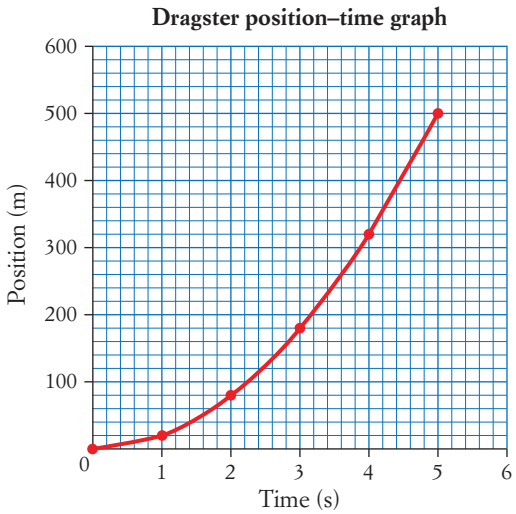
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Answers

1 a

t	0	1	2	3	4	5
x (m)	0	20	80	180	320	500

b

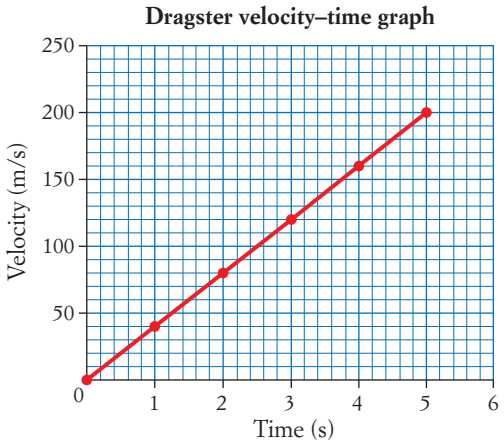


c $v = 40t$

d

t	0	1	2	3	4	5
v (m/s)	0	40	80	120	160	200

e



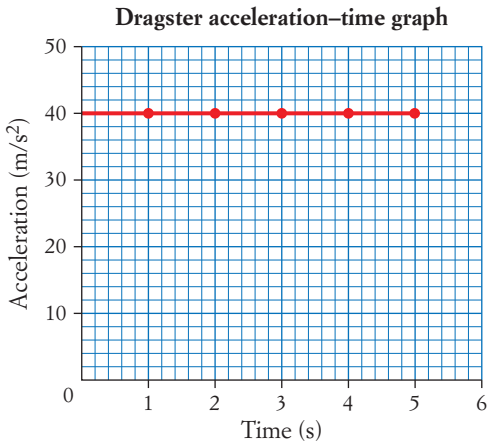
f 40

g

t	0	1	2	3	4	5
a (m/s^2)	40	40	40	40	40	40

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h



2 **a** $v = 4t$

b 8 m/s

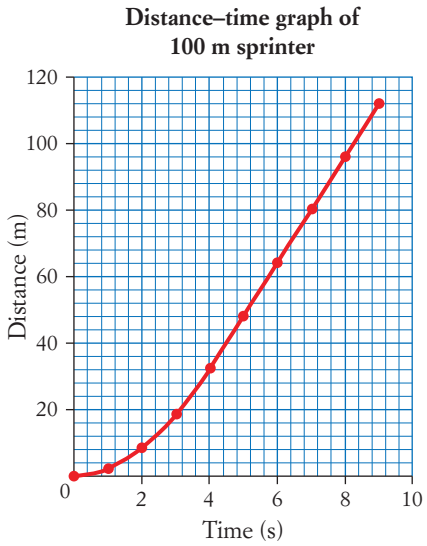
c 16 m/s

d $a = 4$

e 4 m/s^2

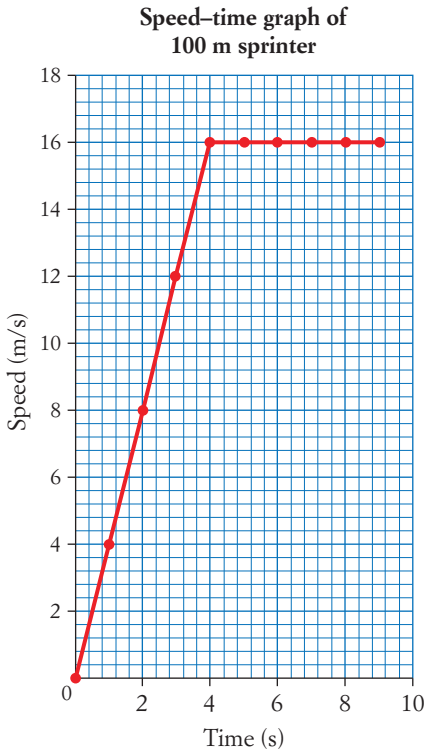
f 0 m/s^2

g

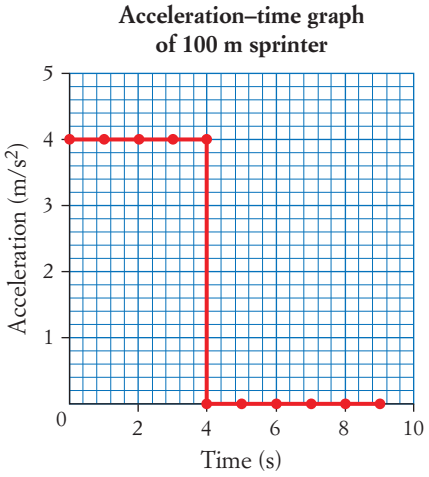


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h



i



3 a $-10t$

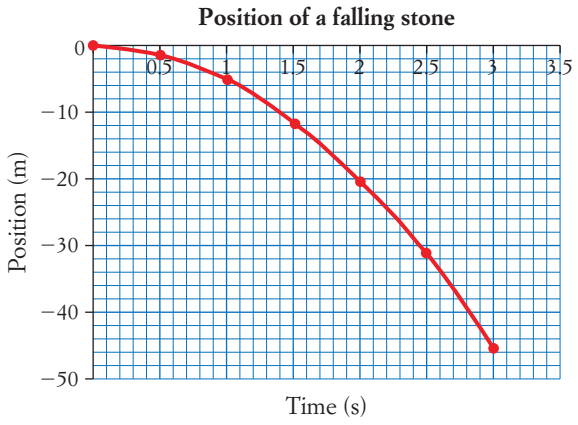
b -10

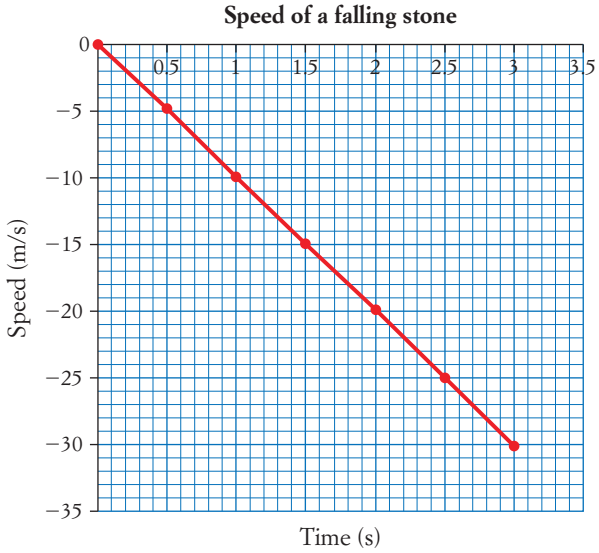
c

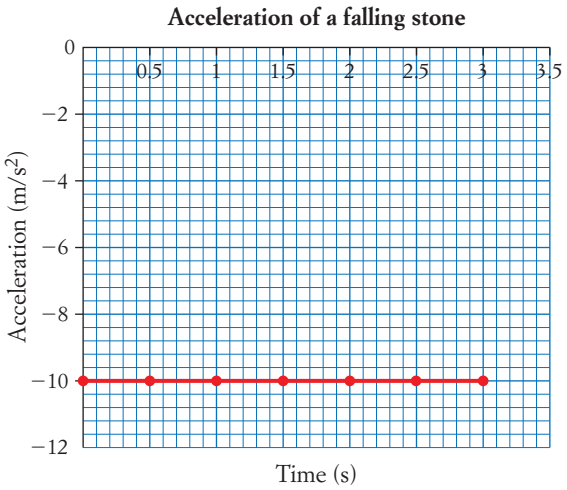
t	0	0.5	1	1.5	2	2.5	3
x (m)	0	-1.25	-5	-11.25	-20	-31.25	-45
v (m/s)	0	-5	-10	-15	-20	-25	-30
a (m/s^2)	-10	-10	-10	-10	-10	-10	-10

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d







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